

Connaissance du Monde

How does the UK's NHS address the challenges of rare disease care ?

Introduction

Imagine waiting 56 years for a diagnosis, only to learn that symptoms dismissed as laziness or depression were actually caused by a rare genetic disorder. This was the reality for one of the Fabry disease patients I interviewed during my work placement at the London Royal Free Hospital (LRFH). To gather these firsthand accounts, I didn't just rely on existing research, I actively reached out to charities, secured a work placement at the LSDU, and conducted in-depth interviews with patients, doctors, nurses, and administrators. This opportunity came after months of persistence: I contacted multiple organisations, proposed a structured research plan, and demonstrated how my academic background in BFI could contribute to their work. Their willingness to collaborate gave me unfiltered access to the realities of rare disease care - from observing consultations to analysing draft NICE guidelines and even writing a scientific review.

Rare diseases affect 5% of the UK population, yet they remain poorly understood and under-resourced. Defined as conditions affecting fewer than 1 in 2,000 individuals, these diseases are often multisystemic, leading to delayed diagnoses (sometimes taking decades) and a lack of specialised treatments. While only 5% of rare diseases currently have approved therapies, none can be fully cured. This raises a critical question, central to my research: How does the UK's NHS address these immense challenges?

This question is directly tied to the ASIBA syllabus through two key strands: "Living in the World" and "Building a Common Future".

Let's begin by examining the NHS's centralised care model—its strengths, its limitations, and the human stories behind the statistics.

I– Centralised Care

In 2021, the NHS announced an ambitious goal: to make the UK a global leader in rare disease care. To achieve this, it implemented a centralised care system through Lysosomal Storage Disorder Units (LSDUs), supported by a Hub-and-Spoke model. During my placement at the Royal Free Hospital's LSDU, I observed how this system functions. The "hub"—a major hospital like the Royal Free—concentrates rare disease specialists, while the "spokes" (local clinics) ensure ERT (Enzyme Replacement Therapy) availability across England. Healthcare professionals (HCPs) in these spokes are trained to administer ERT, but specialists are only available at the hubs. This means therapies are technically accessible regardless of where you live in England—but expert consultations, complex cases, and multidisciplinary care still require travel to LSDUs.

One of the greatest strengths of this centralised approach is its multidisciplinary care. Patients benefit from a team of cardiologists, neurologists, geneticists, and even psychologists, all working together to address the complexity of their condition. Niamh, the lead nurse I interviewed, showed me how this approach not only ensures rigorous medical monitoring but also empowers patients through initiatives like self-infusion training. This program teaches patients to administer their own treatments at home, giving them greater independence and control over their health. Additionally, to ensure consistency across all centres, the NHS relies on standardised protocols, such as NICE guidelines, developed through expert consensus (Delphi panels). Uma even let me review drafts of new guidelines, highlighting the meticulous effort required to provide safe and equitable care for all patients.

While ERT is available nationwide, seeing a specialist still requires traveling to an LSDU—all of which are located in England. Patients from Scotland, Wales, or Northern Ireland face significant disparities, but even within England, the burden is heavy. Maninder, the unit's administrator, explained the logistical challenges: patients often need multiple days off work for appointments, and booking slots is difficult due to high demand. "Some patients travel for hours just for a 30-minute consultation," he told me, highlighting the economic and personal costs of centralisation.

Another persistent problem is diagnostic delays. Despite progress, the average time to diagnose a disease like Fabry remains 10 years, and for rarer conditions, it can take decades. During my placement, I met patients who had seen dozens of doctors before receiving an accurate diagnosis. These delays aren't just a matter of individual suffering; they also have huge human and economic costs, as untreated rare diseases can lead to irreversible complications.

Finally, while centralisation improves access to specialised care, it doesn't fully address equity. Patients in rural or remote areas remain at a disadvantage, and non-white patients are significantly underrepresented in these specialised services. This disparity raises concerns about accessibility and cultural sensitivity within the system. These observations led me to ask: Can a system this centralised truly ensure equitable care for all, or does it risk exacerbating existing inequalities?

II- Innovation

To complement its centralised care model, the NHS is heavily investing in innovation, positioning itself at the forefront of rare disease research and treatment. During my placement at the LSDU, Simon, the unit's pharmacist, walked me through the three major therapeutic advances that have transformed patient outcomes in recent years. First, Enzyme Replacement Therapy (ERT), introduced in 2001, marked a turning point. For diseases like Fabry, ERT slows progression and alleviates symptoms, though it's not a cure. Then came chaperone therapy in 2016 (e.g., migalastat), which targets specific genetic mutations, offering a more precision-based approach. But the most promising - yet challenging - innovation is gene therapy, which aims to replace defective genes entirely. "This could be a game-changer," Simon told me, "but it's not a one-size-fits-all solution." Different mutations respond differently, and not all patients qualify for these cutting-edge treatments.

Beyond therapies, the NHS is exploring wearable tech for real-time patient monitoring and telemedicine to reduce travel burdens. However, as Simon noted, "Funding is the biggest hurdle." ERT alone costs £200,000 per patient per year, and gene therapy could reach £1 million per treatment - unsustainable given the NHS's budget constraints. Currently, 12% of the NHS budget funds highly specialised services for just 5% of the population, leaving little room for expansion.

My interviews also revealed systemic gaps in holistic care. Only 20% of LSDUs have dedicated psychologists, despite the emotional toll of rare diseases. Sophie, from the MPS Society charity, explained how they often step in to fill these gaps - providing support groups, counseling, and even medical equipment the NHS can't supply. "We're a lifeline," she said, "but we shouldn't have to be." This collaboration between charities and the NHS highlights both the system's limitations and its commitment to improvement.

Yet, the most striking tension is between innovation and accessibility. While gene therapy offers hope, its exorbitant cost and limited eligibility raise ethical questions: Should the NHS prioritise cutting-edge treatments for a few, or equitable care for all? The answer, as my fieldwork showed, isn't simple - but it's clear that without structural reforms, even the most advanced innovations won't bridge the equity gap.

IV– Conclusion

My fieldwork at the LSDU and interviews with patients, clinicians, and administrators revealed a healthcare system at a crossroads. The NHS's centralised care model ensures that therapies like ERT are available across England, while its focus on innovation- from chaperone therapy to gene therapy - offers hope for groundbreaking treatments. Yet, these advances are undermined by systemic barriers: geographic disparities, diagnostic delays, and the stark reality that only 5% of rare diseases have approved treatments.

The human cost of these gaps became clear during my placement. I met patients who had waited decades for a diagnosis, others who struggled with the financial and logistical burdens of traveling to LSDUs, and many who felt left behind by a system that prioritises innovation over equity. The NHS's collaboration with charities like the MPS Society- while commendable - highlights a failure to provide holistic care within its own structures.

To bridge these gaps, three structural reforms are urgent:

1. Policy Dynamism: Mandate neonatal screening for detectable rare diseases and GP training modules to reduce diagnostic delays.
2. Sustainable Funding: Shift from pharma dependency to NHS-led clinical trials to lower drug costs and ensure affordability.
3. Holistic Integration: Expand psychological support and charity partnerships to address unmet needs, recognising that rare disease care extends beyond medicine.

As one LSDU consultant told me: "The disease does not manage you. You manage the disease. Dr. Ramaswami" Achieving this requires more than scientific progress - it demands political will, equitable resource allocation, and a commitment to listening to patients' voices. The question now is whether the NHS can reconcile its ambition with the realities of those it serves.

This immersion taught me that rare disease care is about more than protocols and textbooks - it's about the human stories, resilience, and unexpected moments that define each patient's journey. Above all, I learned that listening deeply and adapting quickly can be just as powerful as medical expertise in making a real difference.

Glossary of Terms

NHS	National Health Service
LSDU	Lysosomal Storage Disorder Unit
ERT	Enzyme Replacement Therapy
NICE guidelines	National Institute for Health and Care Excellence
GP; HCP	General practitioner; Health Care Professional
MPS Society	A UK charity supporting patients with mucopolysaccharidosis (MPS) and related rare diseases.
Holistic Care	An approach to healthcare that addresses not just medical needs but also psychological, social, and economic factors affecting patients.